

HOULTON, ME, USA

WMO: 727033

Lat: 46.119N

Lon: 67.793W

Elev: 145

StdP: 99.59

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14609

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-22.7	-30.3	0.2	-24.8	-27.4	0.3	-21.8	10.1	-4.9	9.2	-6.2	1.8	200	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	29.6	20.9	27.8	19.8	26.2	18.9	22.3	27.1	21.4	25.6	20.4	24.3	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.8	24.4	19.9	14.9	23.4	18.9	14.0	22.5	66.2	27.1	62.7	25.7	59.3	24.2	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	8.1	7.3	DB	-30.9	32.5	3.3	1.4	-33.2	33.5	-35.2	34.3	-37.0	35.1	-39.4	36.1	(4)
(5)			WB	-30.9	24.3	3.0	0.8	-33.1	24.9	-34.9	25.4	-36.6	25.8	-38.8	26.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-10.4	-9.1	-3.4	3.8	10.8	15.8	19.2	18.0	13.6	7.1	0.7	-6.0	(6)
(7)		DBStd	11.32	7.17	6.54	5.81	4.18	4.02	3.85	3.01	3.15	4.25	4.36	5.22	6.16	(7)
(8)		HDD10.0	2713	632	534	416	191	38	2	0	0	13	112	281	496	(8)
(9)		HDD18.3	4952	890	767	673	437	237	96	26	44	151	348	529	754	(9)
(10)		CDD10.0	923	0	0	1	4	62	176	284	249	122	23	2	0	(10)
(11)		CDD18.3	119	0	0	0	0	3	20	52	35	10	0	0	0	(11)
(12)		CDH23.3	1213	0	0	1	2	61	226	496	339	87	3	0	0	(12)
(13)		CDH26.7	281	0	0	0	0	13	54	116	78	20	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.4	3.5	3.8	3.6	3.4	3.0	2.7	2.5	2.7	3.1	3.3	3.3	(14)
(15)	Precipitation	PrecAvg	982	68	56	64	66	81	87	91	101	91	88	97	86	(15)
(16)		PrecMax	1426	152	100	124	155	160	234	169	248	250	201	216	148	(16)
(17)		PrecMin	552	6	10	11	11	11	30	40	12	34	28	42	25	(17)
(18)		PrecStd	168	37	22	25	33	38	46	32	54	45	47	40	36	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	8.6	13.8	21.1	28.4	30.9	31.5	31.3	29.3	23.1	16.5	10.9	(19)
(20)			MCWB	9.4	6.6	7.7	10.8	18.0	21.8	22.0	21.5	21.5	16.3	14.8	9.6	(20)
(21)		2%	DB	5.7	5.1	8.9	17.0	24.8	28.1	29.4	28.8	25.8	19.2	13.7	7.2	(21)
(22)			MCWB	4.5	3.1	4.6	9.4	15.1	19.6	21.2	20.6	19.6	14.7	11.5	6.0	(22)
(23)		5%	DB	2.6	2.8	6.7	14.0	22.0	26.0	27.8	26.9	23.2	17.0	11.1	4.4	(23)
(24)			MCWB	1.3	1.2	3.5	8.3	13.5	18.3	20.0	19.5	17.9	13.9	9.1	3.1	(24)
(25)		10%	DB	0.2	1.1	4.6	11.4	19.2	23.8	26.2	25.1	21.3	14.9	8.6	2.4	(25)
(26)			MCWB	-0.8	-0.2	1.8	6.8	12.5	17.1	19.3	18.6	17.1	11.8	6.9	1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	7.5	9.2	14.1	19.7	23.1	23.8	23.2	22.6	18.1	15.5	9.7	(27)
(28)			MCDWB	10.2	8.0	11.4	17.9	25.9	29.0	28.7	28.1	27.6	19.7	16.5	10.4	(28)
(29)		2%	WB	4.6	3.7	6.1	11.4	16.8	21.3	22.4	21.9	20.7	16.1	12.3	6.2	(29)
(30)			MCDWB	5.6	4.7	7.9	14.3	21.7	25.9	27.2	26.3	23.4	18.6	13.2	7.0	(30)
(31)		5%	WB	1.3	1.3	3.7	9.2	15.2	20.0	21.4	20.9	19.4	14.2	9.2	3.4	(31)
(32)			MCDWB	2.3	2.5	6.1	13.1	19.5	23.9	25.8	24.7	22.2	17.1	10.5	4.3	(32)
(33)		10%	WB	-0.8	-0.1	2.1	7.2	13.7	18.5	20.5	19.9	17.8	12.1	6.9	1.3	(33)
(34)			MCDWB	0.2	1.1	4.2	10.9	18.2	22.2	24.5	23.5	20.4	14.4	8.5	2.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	11.6	10.7	10.9	12.6	12.4	12.1	12.6	12.4	10.6	8.3	8.9	(35)
(36)			MCDBR	11.7	10.6	12.6	16.4	17.8	15.5	14.0	14.8	14.0	13.6	11.0	9.6	(36)
(37)			MCWBR	11.4	9.0	9.4	10.1	9.5	8.4	6.9	7.5	8.4	9.3	9.5	9.0	(37)
(38)		5% WB	MCDBR	11.9	9.7	11.2	13.4	13.8	12.8	11.8	11.9	11.7	11.7	10.0	9.5	(38)
(39)			MCWBR	11.7	8.9	9.0	9.8	8.4	7.7	6.4	6.8	7.8	9.1	9.6	9.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.259	0.278	0.310	0.342	0.370	0.391	0.416	0.394	0.364	0.322	0.300	0.278	(40)	
(41)		taud	2.426	2.338	2.366	2.408	2.377	2.317	2.271	2.336	2.400	2.524	2.536	2.453	(41)	
(42)		Ebn at Noon	861	905	920	916	897	875	847	853	850	846	805	792	(42)	
(43)		Edh at Noon	68	91	103	109	117	125	129	117	100	76	63	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.47	2.35	3.52	4.41	5.00	5.41	5.47	4.84	3.90	2.38	1.39	1.07	(44)	
(45)		RadStd	0.17	0.23	0.24	0.41	0.56	0.39	0.45	0.30	0.29	0.22	0.16	0.10	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	+1.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (22 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page